Wanderchat: An AI-Powered Travel Assistant for Personalized Itinerary Planning

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ABSTRACT

Tourism is a broad area that encompasses a number of different activities, all of which require careful planning before leaving home. When students plan their travel outside their home country or state, they are generally overwhelmed by the amount and variety of information available, especially since they typically have limited funds with which to work. The intention of this research is to create a smart travel companion called Wanderchat that will assist users in automating and personalizing their travel through the use of novel techniques such as Natural Language Processing (NLP) to allow the user to communicate with the product in real-time; analysis of the sentiments provided in user reviews using NLP; geographically clustering the identified tourist attractions using K-means clustering; providing the user with personalized recommendations of what to see while using K-Nearest Neighbors (KNN) algorithms; using a computer vision system, such as VGG-16, to identify tourist landmarks in pictures taken by users; and optimizing travel route(s) to minimize travel time using the 2-opt algorithm. Results from testing to-date show that the use of Wanderchat will help students plan their travels with speed and accuracy, while also increasing student satisfaction and reducing the amount of effort and time needed to plan travel. This study also contributes to the development of intelligent systems in the tourism industry by demonstrating a practical implementation of such a system.

.**KEYWORDS :**Travel companion, Smart travel assistant, Trip planner, Travel guide app, AI travel assistant, Journey planner, Travel buddy, Vacation planner, Travel organizer

1 INTRODUCTION

The tourism sector has been altered by the rapid advancement of Artificial Intelligence (AI) from "search engines" and stale data to "dialogue" and dynamic planning with the assistance of AI. Despite the growing accessibility of general-use AI tools, the demand from students for specialized AI systems designed for them is not yet being fully met. When travelling as a student, it can be challenging and stressful to manage numerous variables, including budget constraints, limited time frames, and the desire for authentic cultural experiences. Consequently, students can spend an inordinate amount of time and effort using traditional methods to plan travel.

In many instances, current AI-based travel tools "hallucinate," meaning they create made-up or unnaturally-priced places. Additionally, when students are travelling in groups, the AI travel solution does not adequately provide the necessary highly personalized information required. Both of these significant limitations are rectified with the assistance of a state-of-the-art RAG (Retrieval-Augmented Generation) and MoA (Mixture of Agents) architecture utilized by Wanderchat. By combining real-time web search (Tavily) with structured vector knowledge (FAISS/ChromaDB), the systems of Wanderchat create an enormous level of factual and up-to-date information upon which a student can make travel decisions. Furthermore, the LLM-as-Judge framework will allow Wanderchat to evaluate and improve the level of travel plans before presenting them to the student to ensure they can meet the required level of quality (high fidelity), feasibility, and financial viability—that is, staying in the student's budget. This research illustrates how a multi-agent, feedback-based approach to improving the personalization of travel through the use of AI can have a transformative impact on the personalised tourism industry and how the next generation of digital-native travellers can benefit from a scalable solution..

2 RELATED WORK

Over the past few years, AI has really changed how we plan trips, thanks to big leaps in how computers understand language, how they suggest things, and how they learn from different types of information. Most studies in this area usually fall into three main groups: first, travel helpers that chat with you using advanced AI models; second, systems that learn to recommend places and activities; and third, planning tools that use all sorts of information and pay attention to what's happening around them. In this part, we'll go over these approaches and point out the gaps that our new system aims to fill.

2.1 LLM-Based Conversational Travel Assistants

Recently, thanks to advances in big AI language models (LLMs), we've started seeing travel helpers that can chat with people just like you and me, creating custom trip plans. For example, some GPT-based planners take what users tell them—like what they like, how much money they want to spend, and how long they'll be gone—and put together full trip details, including where to stay, how to get around, and what to do.

These systems usually have parts that help them understand what you're saying, remember what you've talked about, and connect to live data to make things even more personal. And because you can chat back and forth with them, they can keep adjusting the plans based on what you tell them.

However, even though they're pretty flexible and easy to use, most of these LLM-based systems rely on just one main AI brain. This can cause problems, like making up facts, not always being accurate, or giving inconsistent results. Plus, these systems often don't have a good way to double-check the trips they suggest before showing them to you.

2.2 Machine Learning-Based Recommendation System

Another big area of research looks at trip planning as a way to recommend things, using older and mixed machine learning methods. These systems use approaches like looking at what similar people liked, what's in the content itself, or finding things close by to give you tailored suggestions based on your profile, what you've liked before, and current information

For example, tools like K-Nearest Neighbors (KNN) and grouping algorithms (like K-Means) are often used to put destinations together and recommend interesting spots. And to figure out how users and items relate, people often use tricks like TF-IDF or cosine similarity.

Also, making a trip plan often turns into a puzzle where you try to find the best way to do things, using smart algorithms like those based on the Traveling Salesman Problem (TSP) or Team Orienteering Problem (TOP) to create really efficient schedules.

These systems are great at figuring out what users like and finding the best routes, but they usually don't let you chat with them or adapt to changes in real-time. What's more, they often work on their own and aren't built into a single, complete travel helper system.

2.3 Multimodal and Context-Aware Travel Planning Systems

Recently, research has looked into putting together different kinds of information—like pictures, videos, and live data about the environment—into travel planning systems. These systems want to make your trip better by adding in details like the weather, traffic, bus or train times, and local events.

Some platforms use computer vision (which is how computers 'see') to recognize landmarks from pictures you upload, while others connect to live data sources to quickly change plans if things shift. Plus, these systems often mix text, pictures, and location data to give you deeper, more engaging travel ideas.

However, even with these steps forward, many of these systems run into problems with how complex they are, how well they can grow, and how hard it is to smoothly combine all those different types of data. What's more, they don't always have strong ways to make sure everything stays consistent and reliable when dealing with so many different kinds of information.

2.4 Integrated Travel Platforms and Emerging Trends

Besides just making trip plans, some research has focused oncreating full-service travel platforms that bring together planning, booking, finding the best prices, and managing loyalty programs. They use smart predictions and process huge amounts of data to suggest affordable travel options and help you spend your money wisely.

This shows a bigger trend: people want all-in-one travel helpers that do more than just recommend things, also helping with money and logistics. But these systems often care more about making money and might not fully consider what a user really needs, like sticking to a tight budget or planning for a group.

2.5 Research Gaps and Motivation

Even though AI has come a long way in planning trips, there are still some problems with current methods:

Too much dependence on single big AI language models, which can make things up or give unreliable answers.

Not enough real-time information used to change plans on the fly.

Missing a good system where different AI parts work together to think things through and check facts.

Not enough attention paid to trips on a tight budget or for students.

No strong way to check the final output or create well-organized responses.

To fix these issues, our new system, WanderChat, uses a mixed approach. It puts together Retrieval-Augmented Generation (RAG), a system where different AI 'agents' work together (MoA), real-time thinking with ReAct, and a way for it to improve itself. This combined method helps make sure the information is more accurate, more personal, and more

3.SYSTEM ARCHITECTURE AND METHODOLOGY

This section presents the complete architecture, scoring This section describes the overall architecture, data flow, and computational techniques employed in the proposed system, *WanderChat*. The system is designed as a multi-stage intelligent pipeline that integrates Retrieval-Augmented Generation (RAG), a Mixture of Agents (MoA) framework, real-time reasoning using the ReAct paradigm, and a self-refinement mechanism for structured output generation. The objective is to generate accurate, personalized, and context-aware travel itineraries while minimizing hallucinations and improving response reliability.

3.1 System Architecture

The WanderChat framework follows a sequential four-stage architecture:

Input Processing and Context Retrieval (RAG)

Response Generation (Mixture of Agents - MoA)

Dynamic Reasoning and Web Augmentation (ReAct Loop)

Output Structuring and Validation (Self-Refine Loop)

Each module is designed to address specific limitations of conventional travel planning systems, such as lack of personalization, static recommendations, and unstructured outputs. The overall workflow ensures that responses are both semantically relevant and practically feasible.

3.2 Input Processing and Retrieval-Augmented Generation

Upon receiving a user query, the system first encodes the input using a transformer-based embedding model to generate dense semantic vector representations. These embeddings capture contextual meaning and enable efficient similarity-based retrieval.

The encoded vectors are then queried against a vector database implemented using FAISS (Facebook AI Similarity

Search), which employs cosine similarity and Hierarchical Navigable Small World (HNSW) indexing for approximate nearest neighbor search. The top-*k* relevant data points are retrieved and used as contextual knowledge for downstream processing.

The adoption of a Retrieval-Augmented Generation (RAG) framework ensures that the generated responses are grounded in relevant external knowledge, thereby reducing hallucinations and improving factual accuracy.

3.3 Response Generation using Mixture of Agents

The retrieved contextual information, along with the original user query, is processed through a Mixture of Agents (MoA) architecture. This approach extends the concept of Mixture of Experts by leveraging multiple large language models (LLMs) for collaborative response generation.

The MoA framework operates in three distinct layers:

Layer 1 (Parallel Proposition): Multiple LLMs independently generate candidate responses using the same input, introducing diversity in reasoning and content generation.

Layer 2 (Cross-Model Refinement): Each model evaluates all candidate responses and synthesizes an improved output by combining the most relevant and accurate components.

Layer 3 (Aggregation): The refined outputs are consolidated into a single coherent response through an aggregation mechanism.

This multi-agent strategy enhances robustness, reduces individual model bias, and improves the completeness of generated itineraries compared to single-model approaches.

3.4 Real-Time Reasoning via ReAct Framework

To incorporate dynamic and time-sensitive information, the system integrates a ReAct (Reasoning + Acting) framework. This module combines chain-of-thought reasoning with external information retrieval.

The reasoning agent evaluates whether the generated response sufficiently satisfies the user query. If additional information is required—such as weather conditions, traffic updates, or real-time availability—the system performs targeted web searches using external APIs.

The newly retrieved information is then integrated into the response, and the reasoning process is repeated for validation. This iterative loop ensures that the final output is context-aware, up-to-date, and practically relevant for real-world travel scenarios.

3.5 Output Structuring and Self-Refinement

The final stage of the pipeline focuses on transforming the generated response into a structured and user-friendly format. A formatting agent organizes the content into logical components such as day-wise itineraries, travel routes, and budget summaries.

To ensure accuracy and consistency, a self-refinement mechanism based on the LLM-as-a-Judge paradigm is employed. The structured output is evaluated against the original response to detect information loss or inconsistencies. If discrepancies are identified, the system performs a refinement step before delivering the final output.

This approach ensures that the response is not only accurate but also easily interpretable by end users.

3.6 Auxiliary Machine Learning Components

In addition to the core pipeline, WanderChat incorporates several supporting machine learning and optimization techniques:

K-Means Clustering: Used for grouping destinations based on geographical proximity.

K-Nearest Neighbors (KNN): Enables personalized recommendations based on user preferences.

Sentiment Analysis: Analyzes user reviews to improve recommendation quality.

Computer Vision (VGG-16): Identifies landmarks from user-uploaded images.

2-Opt Algorithm: Optimizes travel routes to minimize distance and travel time.

4.SYSTEM DEMONSTRATION

This part explains how the whole system is designed and shows you how our WanderChat system actually works. We built it as a smart travel helper, powered by AI, with different parts that all work together to create trip plans that are really personal and consider what's happening at the moment.

4.1 System Architecture

WanderChat is built in layers, with four main parts: there's the part you interact with, the backend application, the AI processing part, and the data storage part. The part you see lets you chat with it, where you can type in your travel questions, what you like, how much you want to spend, and how long you're going for. Behind the scenes, the backend takes care of your requests, keeps track of your conversation, and makes sure all the different pieces communicate properly.

The real brains of the operation are in the AI processing layer. This part works through a series of steps, which include something called Retrieval-Augmented Generation (RAG), a Mixture of Agents (MoA), ReAct-based thinking, and a Self-Refine feature. The RAG part takes what you ask, turns it into a code the computer understands, and pulls out useful background info from a special database. The MoA system creates several possible answers using different AI language models at the same time. It then checks these answers against each other to make them better and combines them into one clear response. The ReAct part makes the answer even

better by adding live information, like the weather, traffic, and what's available, by searching the web. And finally, the Self-Refine part organizes and checks the final answer to make sure it's clear and makes sense.

The data storage part holds special databases, what users prefer, and connections to other online services to get up-to-the-minute information. In short, here's how the whole process goes:

User Query → Embedding → Context Retrieval (RAG) → Multi-Agent Generation (MoA) → Real-Time Reasoning (ReAct) → Output Structuring (Self-Refine) → Final Itinerary

You can see how everything fits together in the diagram on page 7, which shows how all the main parts work together.

4.2 System Demonstration

To show you what the system can do, let's look at an example request: "Plan a 3-day budget trip to Goa with beaches and nightlife." The system first understands your request and then finds useful travel information by looking for similar patterns in its data. Then, the MoA system creates several possible trip plans. It fine-tunes them and combines them into one clear answer.

Next, the ReAct part checks if it needs any more live information, and if it does, it pulls in fresh details like the weather, how bad the traffic is, or what events are happening locally. The final trip plan is put together in a way that's easy for you to use, with plans for each day, suggested places to visit, the best ways to get around, and an estimated cost.

On top of that, the system can handle different kinds of input. For example, it can recognize famous places from pictures using its computer vision smarts, which helps it give you even better suggestions. Before showing you the final result, it uses a self-checking process to make sure everything is accurate and easy to read.

This example really shows how the system can create trip plans that are flexible, personal, and dependable, all while working quickly and focusing on what you need.surface precisely the signals that drove the system's assessment.

EVALUATION

5.1 Evaluation Objectives

We wanted to see how well the proposed WanderChat system works at creating travel plans that are accurate, unique to each person, and quick. Our main goals were to check if the suggestions were useful, how much they matched what users liked, how fast the system responded, and if people were generally happy with it. These things are really important for knowing if an AI travel helper can actually be used and trusted in real life.

5.2 Experimental Setup

We tested the system with all sorts of travel questions, looking at different places people wanted to go, how much money they had, how long their trips would be, and what they were interested in. We ran these tests in a controlled setting, using regular computer power. The information we used covered things like places to visit, what other users thought, and background details. It also pulled in live data, like the weather and traffic updates. We did these tests many times over to make sure the results were steady, reliable, and didn't just happen by chance.

5.3 Evaluation Metrics and Results

We looked at a few different things to see how well the system performed. First, we checked how accurate the suggestions were, meaning if the places and activities it recommended really matched what users wanted. Then, a personalization score showed how good the system was at tailoring its plans to fit things like a person's budget or interests. We also timed how quickly it could create a plan. And finally, we gathered user feedback on how easy it was to use, how clear the plans were, and if they found them helpful.

What we found was that WanderChat works well across all the things we measured. By combining Retrieval-Augmented Generation with something called the Mixture of Agents framework, the system can put together travel plans that are really relevant and smart about the situation. It gets much better at making personalized plans by using machine learning methods like K-Nearest Neighbors and clustering. Also, a part of the system called the ReAct-based reasoning module makes the suggestions more dependable because it uses live data. Plus, a "Self-Refine" feature makes sure the answers are well-organized, correct, and easy to get. The system is also pretty quick, which means it's good for things where you need answers right away.

5.4 Discussion

Compared to older ways of planning trips or even simpler AI tools, WanderChat gives better and more realistic suggestions. This is because it uses a setup that pulls information from various sources. The way it uses multiple "agents" helps make sure the answers are complete and don't have conflicting information. And because it can adapt to things happening right now, it's more practical to use. On top of that, the way it structures the information makes it much nicer for users. However, how well the system works can change based on if it can get outside information and how good that information is. For the future, we might do bigger studies with more users, add even more live data, and work on making it respond .

5 LIMITATIONS AND FUTURE WORK

a. Current Limitations

6.1 Limitations

Even though WanderChat works pretty well, it still has some issues. First, how well WanderChat works really depends on getting good, up-to-date information from outside sources, like other websites, travel databases, and live updates. If that information isn't complete or is old, the travel plans it creates might not be correct or dependable. Second, because it uses complex AI models and many different parts working together, it takes a lot of computing power. This can make it slow to respond, especially when lots of people are using it or if the computers running it aren't very powerful.

Also, the system mostly works for questions asked in English, so it's not as useful for people who speak other languages or want to ask about specific regions. Since it uses models trained on existing data, sometimes it might give answers that don't quite

make sense or are just made up, even though we try to check facts and pull information from other sources. What's more, while it tries to personalize things for you, it might not fully understand really detailed or complicated preferences, like your feelings about a place, any special needs you have, or things that change very quickly.

6.2 Future Work:

In the future, we plan to make WanderChat better by fixing these issues and adding more features. One big step is to make it work in many languages so more people can use it. If we include information specific to different regions and local knowledge, the recommendations will be even more accurate and culturally appropriate.

We also want to make the system run faster and smoother by choosing better models, saving frequently used info, and spreading the work across many computers, all to make it respond quicker. Using smarter ways to plan things, like AI that learns from experience or planning that juggles several goals at once, can make travel plans even better by finding a good balance between things like cost, how long it takes, and what you prefer.

Later on, we might also connect more deeply with booking websites, payment systems, and live travel services to offer complete travel help from start to finish. Also, adding ways to protect your privacy and let you control your own data will make the system more trustworthy and secure. And finally, we'll do big studies with lots of users and test it in real-world situations to make sure it works well and to keep making the user experience better and better.

CONCLUSION

We've developed WanderChat, an AI travel assistant that makes personalized and efficient travel plans, always keeping your situation in mind. This system uses some clever methods like Retrieval-Augmented Generation (RAG), a setup called a Mixture of Agents (MoA), ReAct-based thinking, and a Self-Refine process. These all help fix the usual issues with older travel planning tools. By blending conversational AI with machine learning and optimization techniques, WanderChat offers a complete way to automatically build your travel itinerary.

The way WanderChat is built helps it give accurate and reliable suggestions, as it pulls information from outside sources. The multi-agent setup also makes the answers better because different parts of the system 'think' together. We've also included real-time data, so your plans stay flexible and are practical for real life. On top of that, it presents information clearly, making the whole thing easier and more pleasant to use. Our tests show that the system performs really well when it comes to giving accurate recommendations, making things personal, and responding quickly.

So, WanderChat helps improve smart tourism by offering a flexible and adaptable way to plan personalized trips. It tackles big problems like having too much information, plans not feeling personal enough, and itineraries that just don't change, especially useful for those watching their budget. Looking ahead, we plan to add support for many languages, connect it with real-time services, and roll it out to a lot more people. This will make it useful in even more situations and have a bigger effect..

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